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QUESTIONS 20 / 20 completed

Factorial

FACTORIAL

Write a Python function to calculate the factorial of a number.

PROGRAM EDITOR

Python

Run

Save

```python
def factorial(n):
    if n < 0:
        raise ValueError("Factorial is not defined for negative numbers.")
    result = 1
    for i in range(2, n + 1):
        result *= i
    return result

n = int(input())
print(factorial(n))
```

OUTPUT

```
120
```

INPUT

5

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QUESTIONS 20 / 20 completed

Max

MAX

Write a Python function to find the maximum of three numbers.

PROGRAM EDITOR

Python

Run Save

```
def max_of_three(a, b, c):
    return max(a, b, c)

a, b, c = map(int, input("Enter a b c: ").split())
print(max_of_three(a, b, c))
```

INPUT

10 25 7

OUTPUT

```
Enter a b c: 25
```

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QUESTIONS 20 / 20 completed

Palindrome

PALINDROME

Write a Python function to check if a string is a palindrome or not.

PROGRAM EDITOR

Python

Run Save

```
def is_palindrome(s: str) -> bool:
    s = s.lower().strip()
    return s == s[::-1]

print(is_palindrome("madam"))
```

OUTPUT

```
True
```

INPUT

madam

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QUESTIONS 20 / 20 completed

Area

AREA

Write a Python function to calculate the area of a triangle.

PROGRAM EDITOR

Python

Run

Save

```
def triangle_area(base, height):
    return 0.5 * base * height

print(triangle_area(10, 5))
```

OUTPUT

```
25.0
```

INPUT

base = 10
height = 5

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QUESTIONS 20 / 20 completed

Count vowels

COUNT VOWELS

Write a Python function to count the number of vowels in a string.

PROGRAM EDITOR

Python

Run Save

```python
def count_vowels(s: str) -> int:
    vowels = set("aeiouAEIOU")
    return sum(1 for ch in s if ch in vowels)

s = input().strip()
print(count_vowels(s))
```

OUTPUT

```
3
```

INPUT

Hello World

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QUESTIONS 20 / 20 completed

Odd or Even

ODD OR EVEN

Write a Python function to check if a number is odd or even.

PROGRAM EDITOR

Python

Run Save

```
def odd_or_even(n):
    if n % 2 == 0:
        return "Even"
    else:
        return "Odd"

num = int(input("Enter a number: "))
print(odd_or_even(num))
```

OUTPUT

```
Enter a number: Odd
```

INPUT

5

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QUESTIONS 20 / 20 completed

Reverse

REVERSE

Write a Python function to reverse a list.

PROGRAM EDITOR

Python

Run Save

```
def reverse_list(lst):
    return lst[::-1]

print(reverse_list([1, 2, 3, 4, 5]))
```

OUTPUT

```
[5, 4, 3, 2, 1]
```

INPUT

1 2 3 4 5

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QUESTIONS 20 / 20 completed

Prime or not

PRIME OR NOT

Write a Python function to check if a number is prime or not.

PROGRAM EDITOR

Python

Run Save

```
def is_prime(n):
    if n <= 1:
        return False
    if n <= 3:
        return True
    if n % 2 == 0 or n % 3 == 0:
        return False

    i = 5
    while i * i <= n:
        if n % i == 0 or n % (i + 2) == 0:
            return False
        i += 6
    return True

num = int(input())

if is_prime(num):
    print("Prime")
else:
    print("Not Prime")
```

OUTPUT

```
Prime
```

INPUT

17

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QUESTIONS 20 / 20 completed

Ascending order

ASCENDING ORDER

Write a Python function to sort a list in ascending order.

PROGRAM EDITOR

Python

Run Save

```
def ascending_sort(arr):
    return sorted(arr)

print(ascending_sort([5, 2, 9, 1]))
```

OUTPUT

```
[1, 2, 5, 9]
```

INPUT

5, 2, 9, 1

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QUESTIONS 20 / 20 completed

Remove

REMOVE

Write a Python function to remove duplicate elements from a list.

PROGRAM EDITOR

Python

Run Save

```
def remove_duplicates(lst):
    seen = set()
    result = []
    for x in lst:
        if x not in seen:
            seen.add(x)
            result.append(x)
    return result

print(remove_duplicates([1, 2, 2, 3, 1, 4]))
```

OUTPUT

```
[1, 2, 3, 4]
```

INPUT

1 2 2 3

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QUESTIONS 20 / 20 completed

Sum

SUM

Write a Python program to calculate the sum of all elements in a list.

PROGRAM EDITOR

Python Run Save

```
numbers = list(map(int, input("Enter elements of the list separated by space: ").split()
total = sum(numbers)
print("Sum =", total)
```

OUTPUT

```
Enter elements of the list separated by space: Sum = 15
```

INPUT

1 2 3 4 5

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QUESTIONS 20 / 20 completed

2nd Max

2ND MAX

Write a Python program to find the second largest element in a list.

PROGRAM EDITOR

Python

Run Save

```
a = list(map(int, input().split()))
s = sorted(set(a))
print(s[-2])
```

OUTPUT

```
9
```

INPUT

10 3 5 9 9 2

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QUESTIONS 20 / 20 completed

GCd

GCD

Write a Python program to find the greatest common divisor (GCD) of two numbers.

PROGRAM EDITOR

Python

Run Save

```python
def gcd(a, b):
    while b != 0:
        a, b = b, a % b
    return abs(a)
num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))

print("GCD =", gcd(num1, num2))
```

OUTPUT

```
Enter first number: Enter second number: GCD = 12
```

INPUT

24
36

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QUESTIONS 20 / 20 completed

LCM

LCM

Write a Python program to find the LCM of two numbers.

PROGRAM EDITOR

Python

Run Save

```python
def gcd(a, b):
    while b != 0:
        a, b = b, a % b
    return a

def lcm(a, b):
    if a == 0 or b == 0:
        return 0
    return abs(a * b) // gcd(a, b)

a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

print("LCM =", lcm(a, b))
```

OUTPUT

```
Enter first number: Enter second number: LCM = 36
```

INPUT

12
18

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QUESTIONS 20 / 20 completed

Newton-Raphson

NEWTON-RAPHSON

Write a Python program to find the square root of a number using the Newton-Raphson method.

PROGRAM EDITOR

Python | Run | Save

```python
a = float(input("Enter a number (a): "))

if a < 0:
    print("Square root is not real for negative numbers.")
else:
    x = a if a != 0 else 0.0

    for _ in range(1000):
        if x == 0:
            break
        x1 = 0.5 * (x + a / x)
        if abs(x1 - x) < 1e-10:
            x = x1
            break
        x = x1

    print("Square root =", x)
```

OUTPUT

```
Enter a number (a): Square root = 3.0
```

INPUT

```
9
```

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QUESTIONS 20 / 20 completed

N natural

N NATURAL

Write a Python program to find the sum of the first N natural numbers.

PROGRAM EDITOR

Python

Run Save

```
n = int(input("Enter N: "))
sum_n = n * (n + 1) // 2
print(sum_n)
```

OUTPUT

```
Enter N: 15
```

INPUT

5

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QUESTIONS 20 / 20 completed

Sum of numbers

SUM OF NUMBERS

Write a Python program to find the sum of digits in a number.

PROGRAM EDITOR

Python

Run Save

```python
n = input("Enter a number: ").strip()

if n.startswith('-'):
    n = n[1:]

total = 0
for ch in n:
    total += int(ch)

print("Sum of digits:", total)
```

OUTPUT

```
Enter a number: Sum of digits: 15
```

INPUT

12345

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QUESTIONS 20 / 20 completed

Palindrome Num.

PALINDROME NUM.

Write a Python program to check if a number is a palindrome or not.

PROGRAM EDITOR

Python

Run Save

```python
n = input("INPUT: Enter a number: ")
s = n.strip()

if s == s[::-1]:
    print("OUTPUT: Palindrome Number")
else:
    print("OUTPUT: Not a Palindrome Number")
```

INPUT

121

OUTPUT

```
INPUT: Enter a number: OUTPUT: Palindrome Number
```

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QUESTIONS 20 / 20 completed

Factorial

FACTORIAL

Write a Python program to find the factorial of a number using a loop.

PROGRAM EDITOR

Python

Run Save

```python
n = int(input("Enter a number: "))

fact = 1
for i in range(1, n + 1):
    fact *= i

print("Factorial of", n, "is:", fact)
```

OUTPUT

```
Enter a number: Factorial of 5 is: 120
```

INPUT

5

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QUESTIONS 20 / 20 completed

Multiplication

MULTIPLICATION

Write a Python program to print the multiplication table of a given number.

PROGRAM EDITOR

Python

Run Save

```python
n = int(input("Enter a number: "))

for i in range(1, 20):
    print(f"{n} x {i} = {n*i}")
```

OUTPUT

```
Enter a number: 3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
```

INPUT

3